

Positive Edge Detector

//Verilog code

```
`timescale 1ns / 1ps
```

```
module pos_edge_detect (
```

```
    input clk,
```

```
    input signal_now,
```

```
    output reg adelay);
```

```
    reg signal_old;
```

```
    reg adelay2;
```

```
    always @(posedge clk) begin
```

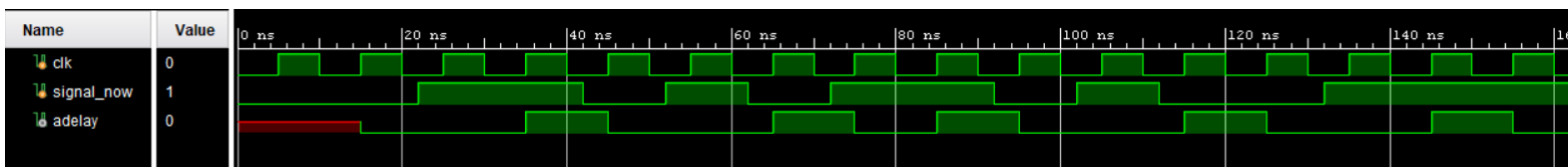
```
        adelay2 <= signal_now & ~signal_old;
```

```
        adelay <= adelay2;
```

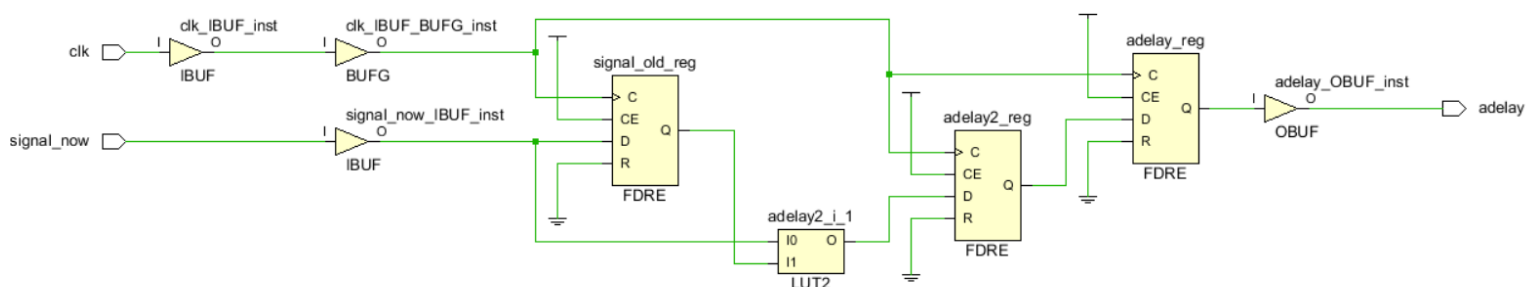
```
        signal_old <= signal_now;
```

```
    end
```

```
endmodule
```



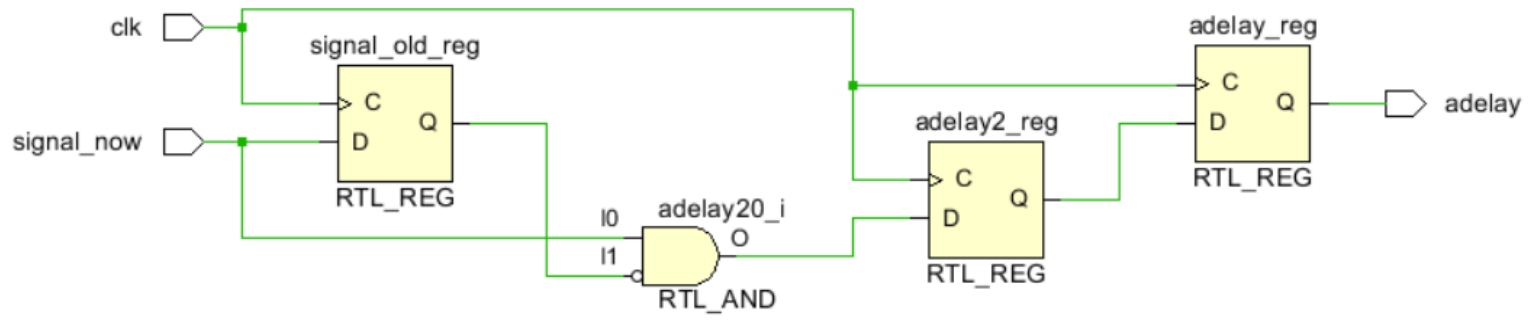
Synthesis Analysis schematic:



Look Up Table:

Cell Properties		
adelay2_i_1		
I1	I0	O=I0 & !I1
0	0	0
0	1	1
1	0	0
1	1	0

RTL Analysis schematic:



Both Edge Detector

//Verilog code

```
`timescale 1ns / 1ps
```

```
module edge_detector (
```

```
    input clk,
```

```
    input signal_now,
```

```
    output reg pos_edge,
```

```
    output reg neg_edge);
```

```
    reg signal_old;
```

```
    always @(posedge clk) begin
```

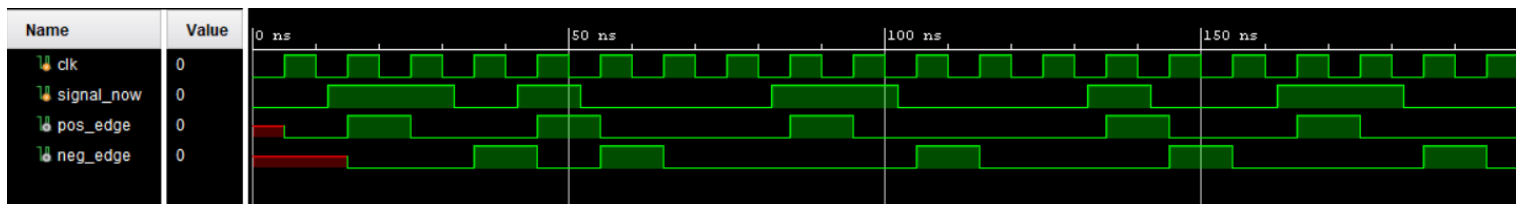
```
        pos_edge <= signal_now & ~signal_old;
```

```
        neg_edge <= ~signal_now & signal_old;
```

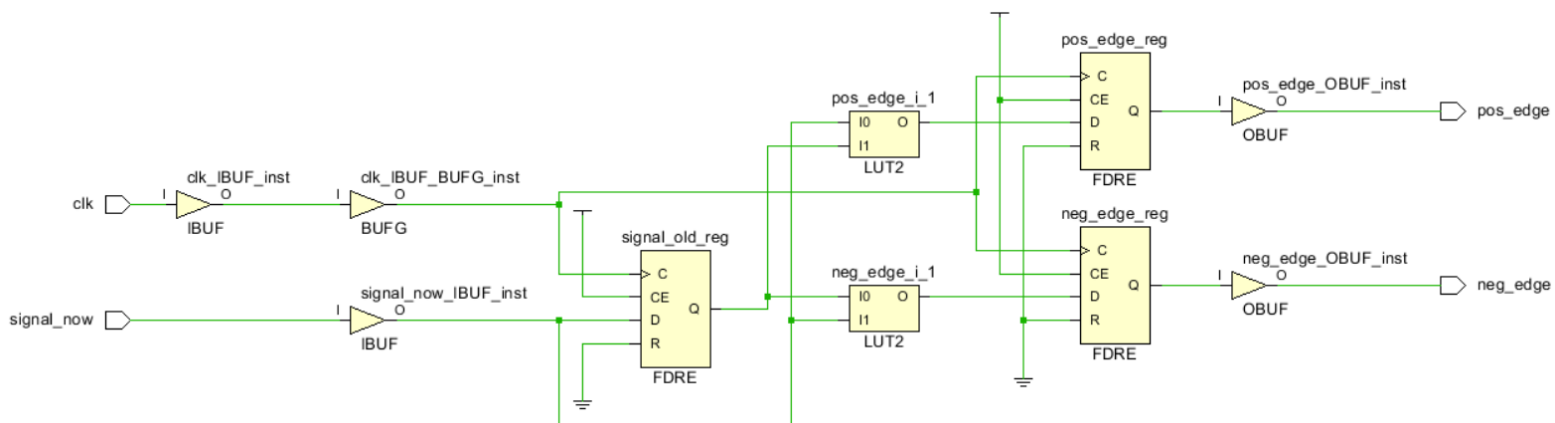
```
        signal_old <= signal_now;
```

```
    end
```

```
endmodule
```



Synthesis analysis schematic:



Look Up Table: LUT2->

Cell Properties		
pos_edge_i_1		
I1	I0	O=I0 & !I1
0	0	0
0	1	1
1	0	0
1	1	0

RTL Analysis schematic:

